

Problem Set 4

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Question 1

Part (a)

```
R2 <- function(y, yhat) {  
  rss <- sum((y - yhat)^2)  
  tss <- sum((y - mean(y))^2)  
  1 - (rss/tss)}
```

Part (b)

```
MSE <- function(y, yhat) {  
  mean((y - yhat)^2)}  
RMSE <- function(y, yhat) {  
  sqrt(MSE(y, yhat))}  
MAE <- function(y, yhat) {  
  mean(abs(y - yhat))}
```

Part (c)

- 1. charges ~ age
- 2. charges ~ age + bmi
- 3. charges ~ age + bmi + smoker
- 4. charges ~ age + bmi + smoker + children
- 5. charges ~ age + sex + bmi + children + smoker + region.

Part (d)

R^2 for models 1–5: 0.08408906 0.1153287 0.7378867 0.7403448 0.7417302

Adjusted R^2 for models 1–5: 0.08323067 0.1136689 0.7371483 0.7393686 0.739781

Part (e)

	Predictors	Validation_MSE	Validation_RMSE
1	age	124222532	11145.516
2	age + bmi	122707454	11077.340
3	age + bmi + smoker	30367055	5510.631
4	age + bmi + smoker + children	30187681	5494.332
5	age + sex + bmi + children + smoker + region	30185004	5494.088

	Validation_MAE
1	8813.605
2	8761.334
3	3941.527
4	3935.529
5	3929.741

Part (f)

The model using age + bmi + smoker + children gives you the best MSE (34948041)

Question 2

The equation from the lm() summary output is:
Price = 18.69496 – 0.07807(Miles) + 9.45872(ModelTacoma) – 11.02341(ModelVolt)

Validation Predictions: 20.76826 20.57308 20.27642 24.67176 24.28921 25.84281 20.50282 20.78387
22.1579 14.92418 16.57926 11.91068 13.65944 17.24286 17.60979 17.46926 0.746741 7.093832 5.571467

RMSE: 3.269519

Question 3

Part (a)

```
NOVA <- read.csv("/Users/aribarua/Desktop/R Docs/NOVA.csv")  
NOVA2 <- NOVA
```

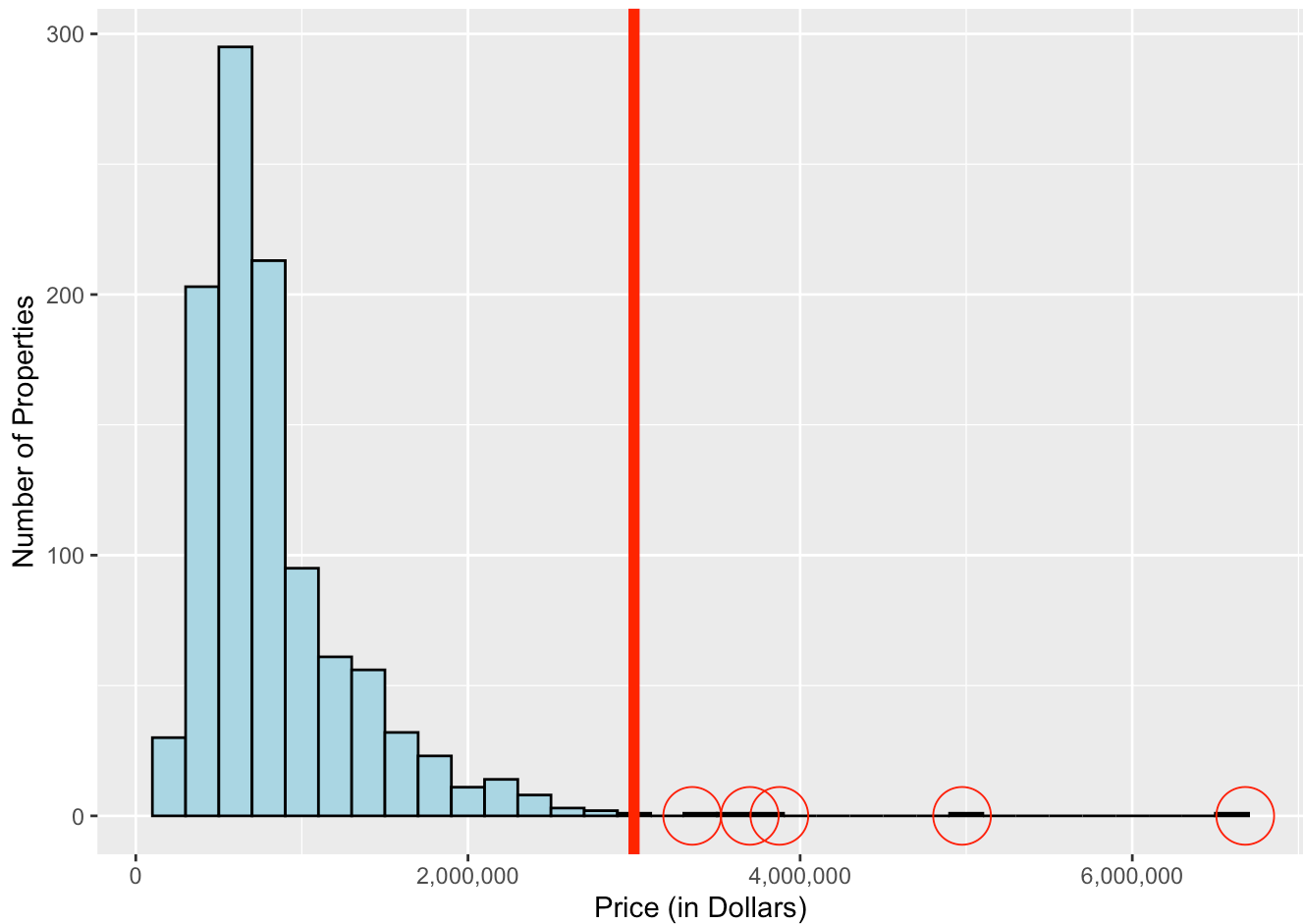
Part (b)

```
table(NOVA2$CountySchool)
```

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Part (c)



Part (d)

	Bedrooms	Bathrooms	LivingSQFT	Age	DistToWH	LotSize	ListingType	Floors
224	5	7	7573	0	7996.266	0.37	Single Family	3
273	5	9	5000	1	5663.079	0.52	Single Family	3
995	7	12	15000	5	24674.127	7.97	Single Family	3
1047	5	7	12000	10	24153.365	2.83	Single Family	3
1052	5	8	8000	2	16714.793	3.39	Single Family	3
	CountySchool		Garage	Price				
224	ARLINGTON COUNTY PUBLIC SCHOOLS		2	3875000				
273	ARLINGTON COUNTY PUBLIC SCHOOLS		3	3350000				
995	FAIRFAX COUNTY PUBLIC SCHOOLS		4	6680000				
1047	FAIRFAX COUNTY PUBLIC SCHOOLS		4	3697000				
1052	FAIRFAX COUNTY PUBLIC SCHOOLS		3	4975000				

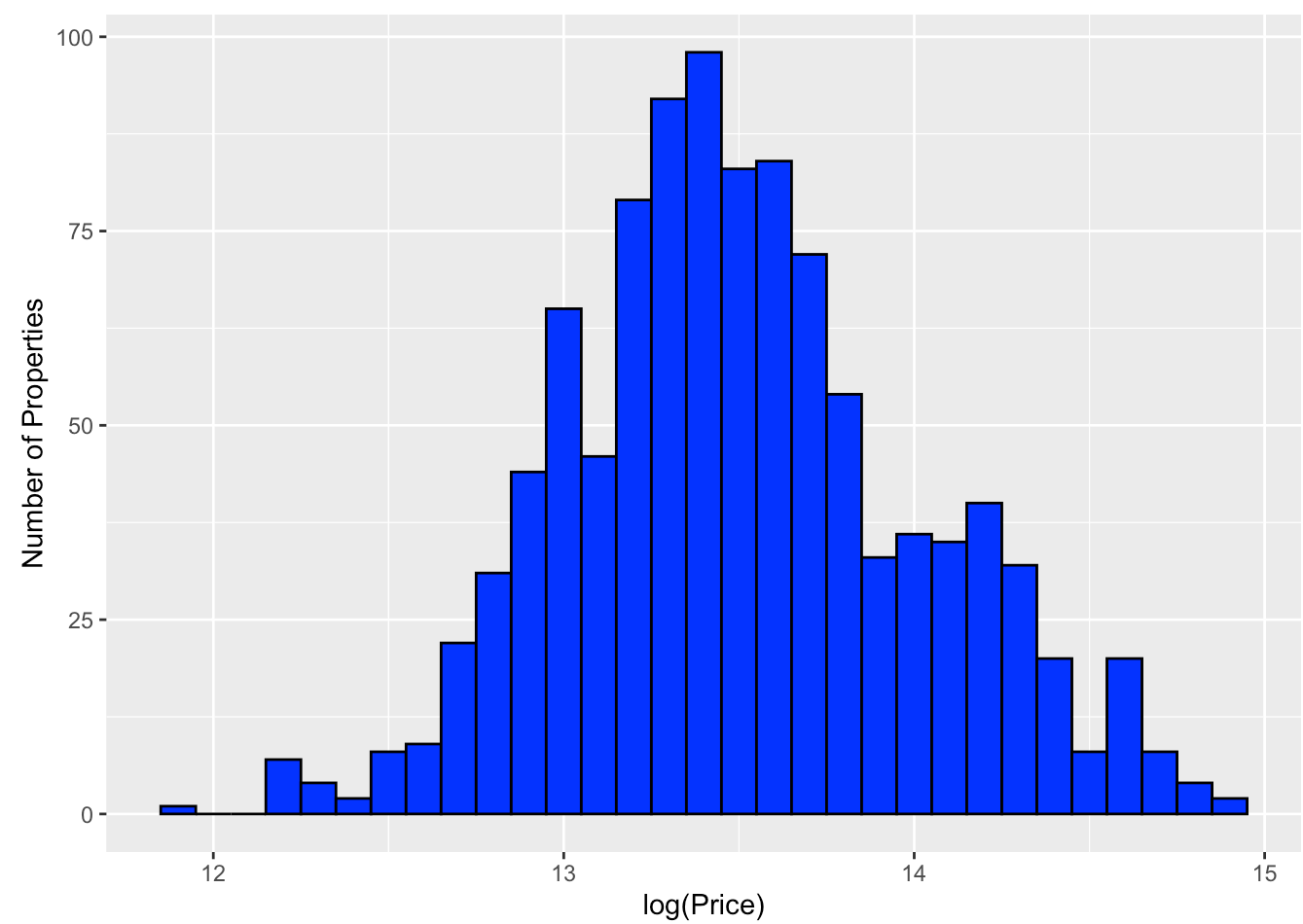
These \$3M+ homes have very large LivingSQFT and/or LotSize, and they are concentrated in Arlington and Fairfax counties.

Part (e)

```
NOVA2 <- NOVA2[NOVA2$Price <= 3000000 & !(NOVA2$CountySchool %in% c("LOUDOUN COUNTY PUBLIC SCHOOLS", "P  
nrow(NOVA2)
```

[1] 1039

Part (f)



This distribution is much more symmetric/normal than before the outliers were removed.

Part (g)

```
N0VA2$Single <- ifelse(N0VA2$ListingType == "Single Family", 1, 0)

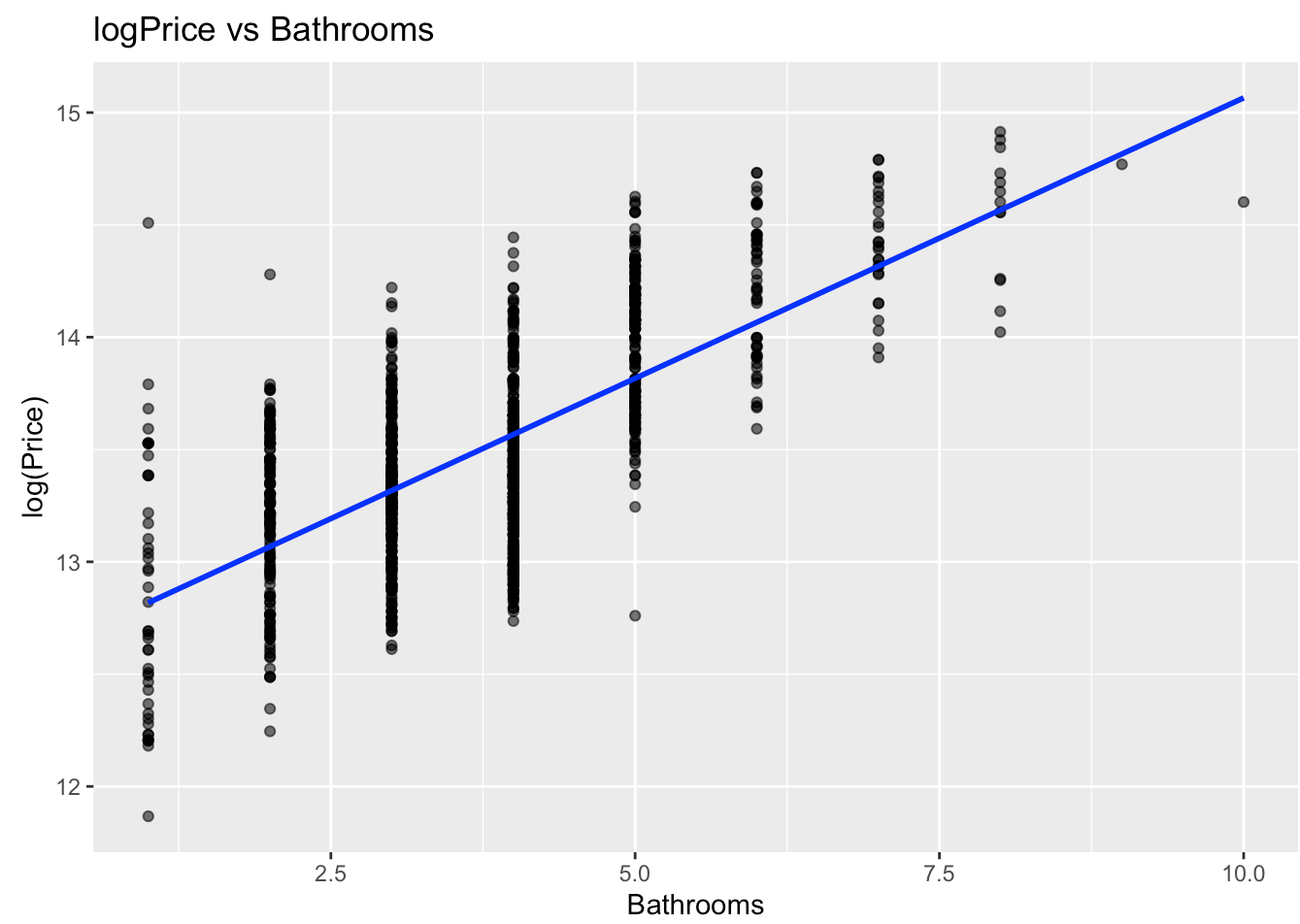
N0VA2$Arlington <- ifelse(N0VA2$CountySchool == "ARLINGTON COUNTY PUBLIC SCHOOLS", 1, 0)
```

Question 4

Part (a)

```
ggplot(N0VA2, aes(x = Bathrooms, y = logPrice)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(title = "logPrice vs Bathrooms",
       x = "Bathrooms",
       y = "log(Price)")
```

`geom\_smooth()` using formula = 'y ~ x'



Yes, there's a pretty clear linear relationship.

Part (b)

```
(cor(N0VA2$logPrice, N0VA2$Bathrooms))
```

[1] 0.6905459

Part (c)

```
model_bath <- lm(logPrice ~ Bathrooms, data = NOVA2)
summary(model_bath)
```

Call:
lm(formula = logPrice ~ Bathrooms, data = NOVA2)

Residuals:

Min	1Q	Median	3Q	Max
-1.05669	-0.25705	-0.01275	0.26123	1.69103

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.567869	0.032509	386.60	<2e-16 ***
Bathrooms	0.249756	0.008124	30.75	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.364 on 1037 degrees of freedom
Multiple R-squared: 0.4769, Adjusted R-squared: 0.4763
F-statistic: 945.2 on 1 and 1037 DF, p-value: < 2.2e-16

$\text{logPrice} = 12.567869 + 0.249756(\text{Bathrooms})$

Part (d)

```
summary(model_bath)$r.squared
```

[1] 0.4768537

About 47.68537% of the variation in logPrice is explained by the number of bathrooms.

Part (e)

```
summary(model_bath)$sigma
```

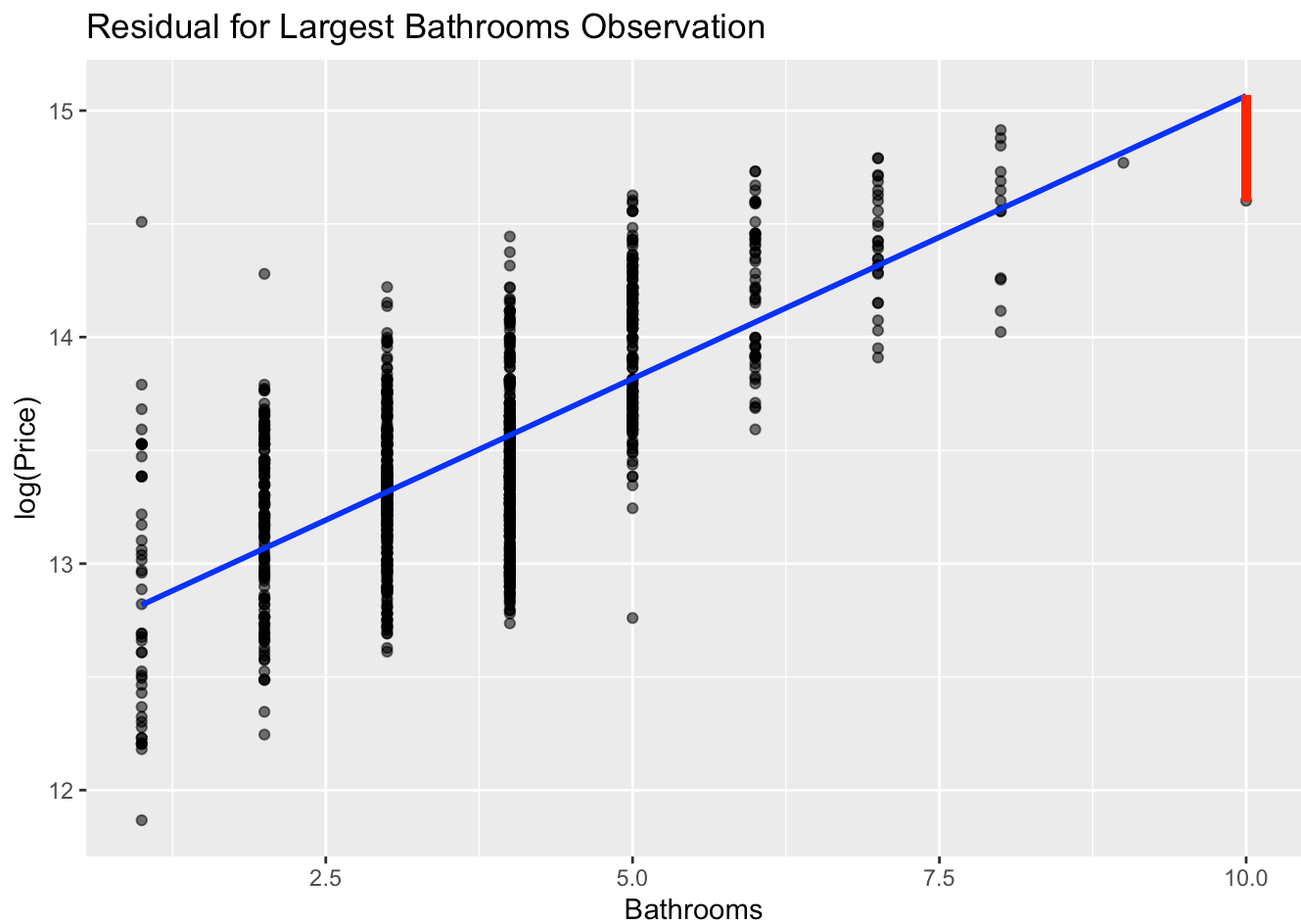
[1] 0.36397

On average, logPrice deviates from the regression line by about 0.36397 units (in log-dollars).

Part (f)

Warning in geom_segment(aes(x = max_obs\$Bathrooms, xend = max_obs\$Bathrooms, : All aesthetics have length 1, but the data has 1039 rows.
i Please consider using `annotate()` or provide this layer with data containing a single row.

`geom\_smooth()` using formula = 'y ~ x'



Question 5

Part (a)

Call:
lm(formula = logPrice ~ Bathrooms + Arlington, data = NOVA2)

Residuals:

Min	1Q	Median	3Q	Max
-1.33365	-0.19347	0.01088	0.18540	1.78656

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.483495	0.026786	466.04	<2e-16 ***
Bathrooms	0.238603	0.006648	35.89	<2e-16 ***
Arlington	0.478650	0.020969	22.83	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.297 on 1036 degrees of freedom
Multiple R-squared: 0.6519, Adjusted R-squared: 0.6512
F-statistic: 970.1 on 2 and 1036 DF, p-value: < 2.2e-16

logPrice = 12.483495 + 0.238603(Bathrooms) + 0.478650(Arlington)

Part (b)

logPrice = 12.483495 + 0.238603(2) + 0.478650(0) = 12.483495 + 0.477206 + 0 = 12.960701

Part (c)

```
(bath_beta <- coef(m_q5) ["Bathrooms"])
```

Bathrooms
0.2386029

Holding Arlington constant, a 1-unit increase in Bathrooms changes expected logPrice by 0.2386029 on average.

Part (d)

```
(arl_beta <- coef(m_q5) ["Arlington"])
```

Arlington
0.4786502

Holding Bathrooms constant, Arlington homes (Arlington=1) have expected logPrice higher by 0.4786502 than Fairfax (Arlington=0).

Part (e)

```
confint(m_q5, parm = "Bathrooms", level = 0.90)
```

	5 %	95 %
Bathrooms	0.2276589	0.2495469

Part (f)

```
r2 <- summary(m_q5)$r.squared  
cat("R^2 =", r2)
```

R^2 = 0.6519163

Part (g)

```
r2_adj <- summary(m_q5)$adj.r.squared  
cat("Adjusted R^2 =", r2_adj, "\n")
```

Adjusted R^2 = 0.6512444